

CS611 Machine Learning Engineering

Lecture 4: Model Development

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Outline

Case Study (1+ hour)

Lecture (1+ hour)

- Data Split

- Model training and evaluation

- Hyperparameter tuning

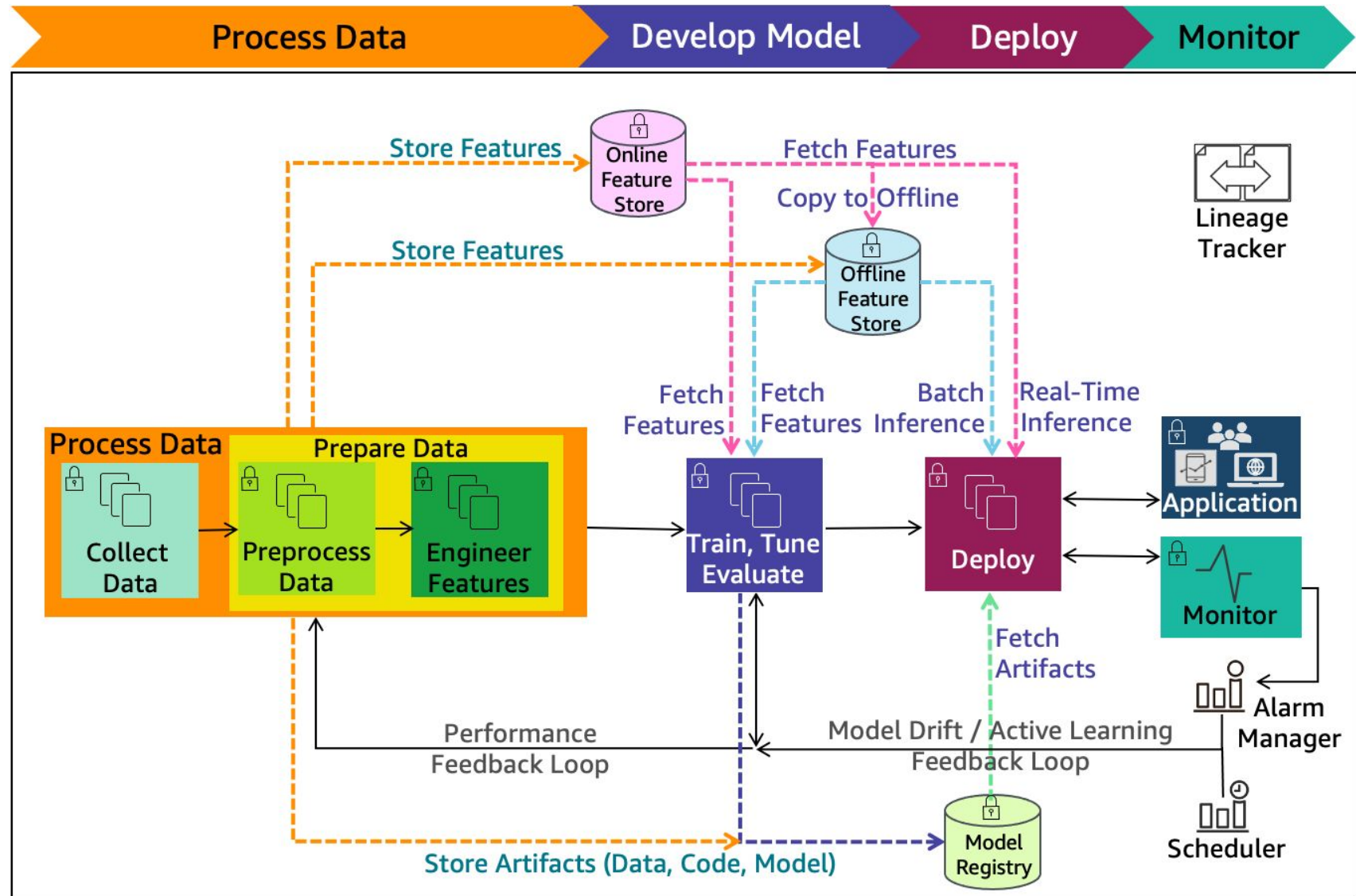
- Calibration

- Distributed training

- Infrastructure: GPU, specialized chip

Lab (20 mins)

Machine Learning Lifecycle



Case Study

You are the Data Scientist at a bank.

You are tasked to build a credit scoring model to predict if a customer will default on a loan.

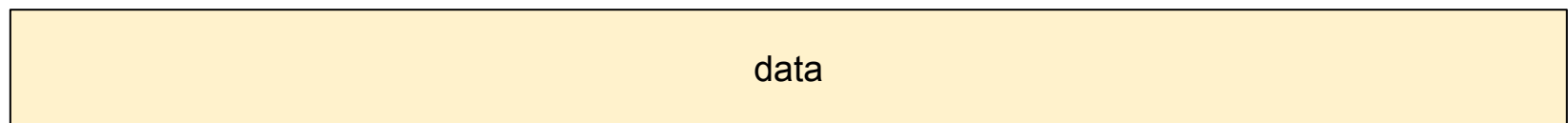
At the moment, you have access to the loan management system which shows daily repayments of loans by customers.

What algo will you try?

You can ask me any questions about the case / business.

Data Split

Class discussion: How do you split data?



1 Jan 2023

1 Jan 2025

Data Split

Training / Validation / Test	OOT 1	OOT 2	OOT 3
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1 Jan 2023

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Training Data: For training models

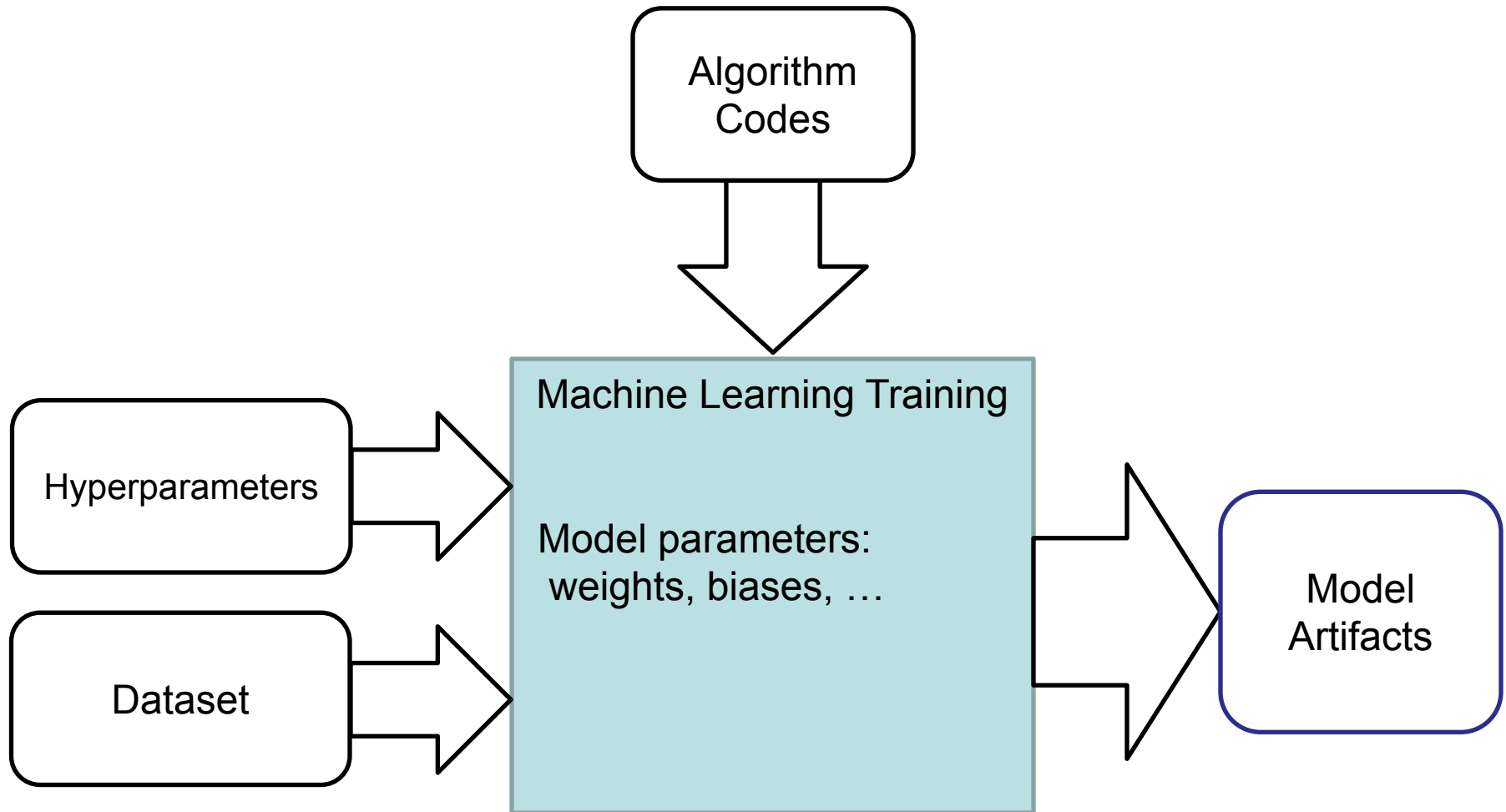
Validation Data: For tuning models

Test Data: For evaluating models

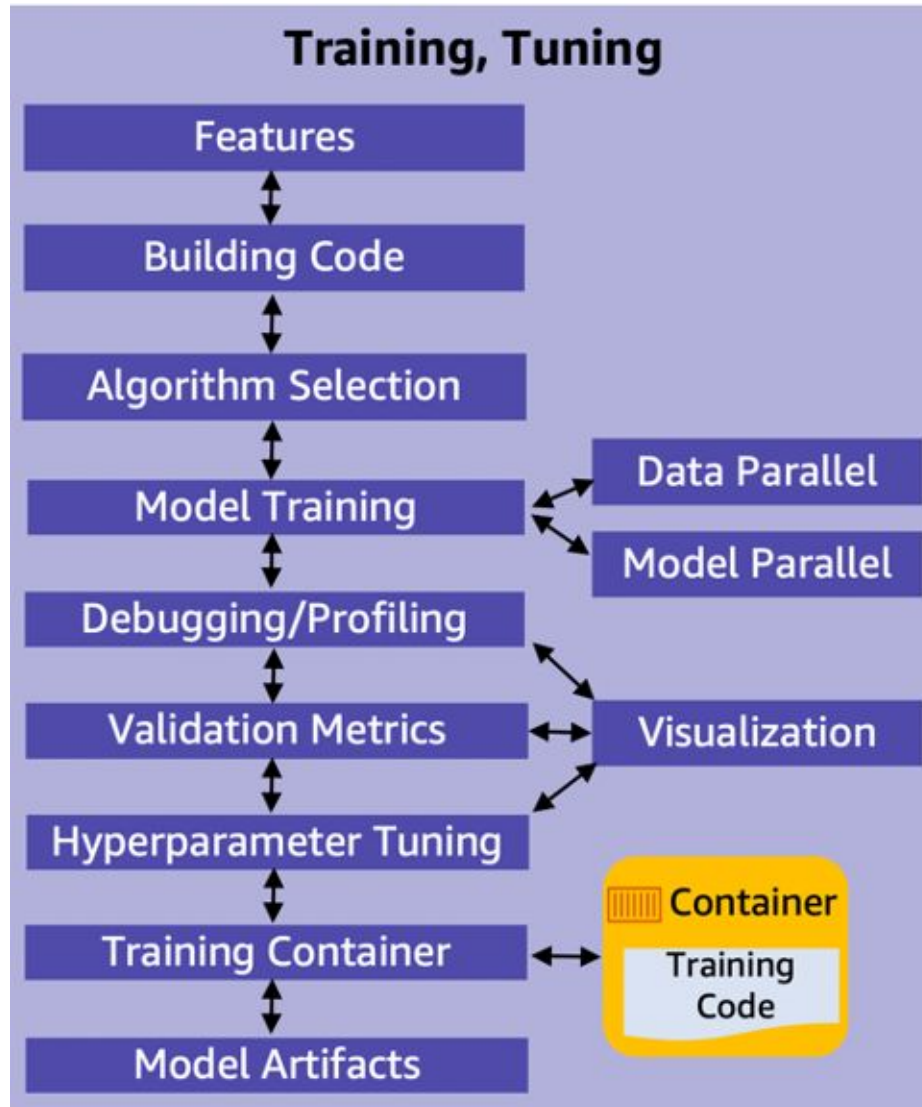
OOT Data: To evaluate model and continuously monitor models out-of-time (OOT)

Model training and evaluation

Machine Learning Model Training



Model Training and Tuning



Class discussion:

How do you evaluate models?

Training / Validation / Test	OOT 1	OOT 2	OOT 3
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1 Jan 2023

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Model Evaluation

Performance Metrics

Check the model type; e.g. classification, regression, clustering, ranking, etc.

Find the appropriate metrics tied to business impact:

<https://scikit-learn.org/stable/api/sklearn.metrics.html#module-sklearn.metrics>

Stability Metrics

Population Stability Index (PSI):

Characteristic Stability Index (CSI):

Training / Validation / Test	OOT 1	OOT 2	OOT 3
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1 Jan 2023

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Hyperparameter Tuning

Hyperparameter Tuning

- Finding the combination of hyperparameters that give the best result.
- Running hyperparameter tuning job requires:
 - Objective metric
 - Hyperparameter ranges
 - Job parameters
- Several strategies: grid search, random search, bayesian optimization, etc.

Calibration

Class Discussion:
**How do you get the probability value
from a model inference?**

Calibration

- Calibrating raw model inference output to a business usable format.
- Probabilistic Calibration:
 - “When performing classification you often want not only to predict the class label, but also obtain a probability of the respective label. This probability gives you some kind of confidence on the prediction. Some models can give you poor estimates of the class probabilities and some even do not support probability prediction”
 - source:
<https://scikit-learn.org/stable/modules/calibration.html>

Distributed Training

Distributed Training

- Two types of challenges in model trainings:
 - **Large datasets** take a long time to train
 - **Large models** (models with many parameters) can be larger than the memory capacity of single GPU
- Distributed training involves multiple machines (nodes) to train large datasets or large models.

Distributed Training

- Two distributed training strategies:
 - Data parallelism:
 - splitting the training set in mini-batches evenly distributed across nodes.
Thus, each node only trains the model on a fraction of the total dataset.
 - More common approach to distributed training.
 - Model parallelism: splitting a model up between multiple devices or nodes.
- Popular deep learning frameworks such as PyTorch, Tensorflow have distributed learning package.

Class discussion:
**What are the pros and cons of model
parallelism vs single model?**

Infrastructure: GPU & Specialized Chips

What is GPU

- Graphics Processing Unit (GPU) is a specialized processor that was originally designed to accelerate 3D graphics rendering.
- Over time it became more flexible and programmable which allowed developers to broaden the horizons and use GPUs for other tasks.
- The greatest strength of a GPU is the ability to process many pieces of data at the same time. This makes GPUs quite useful devices for machine learning (ML).



How GPU Help ?



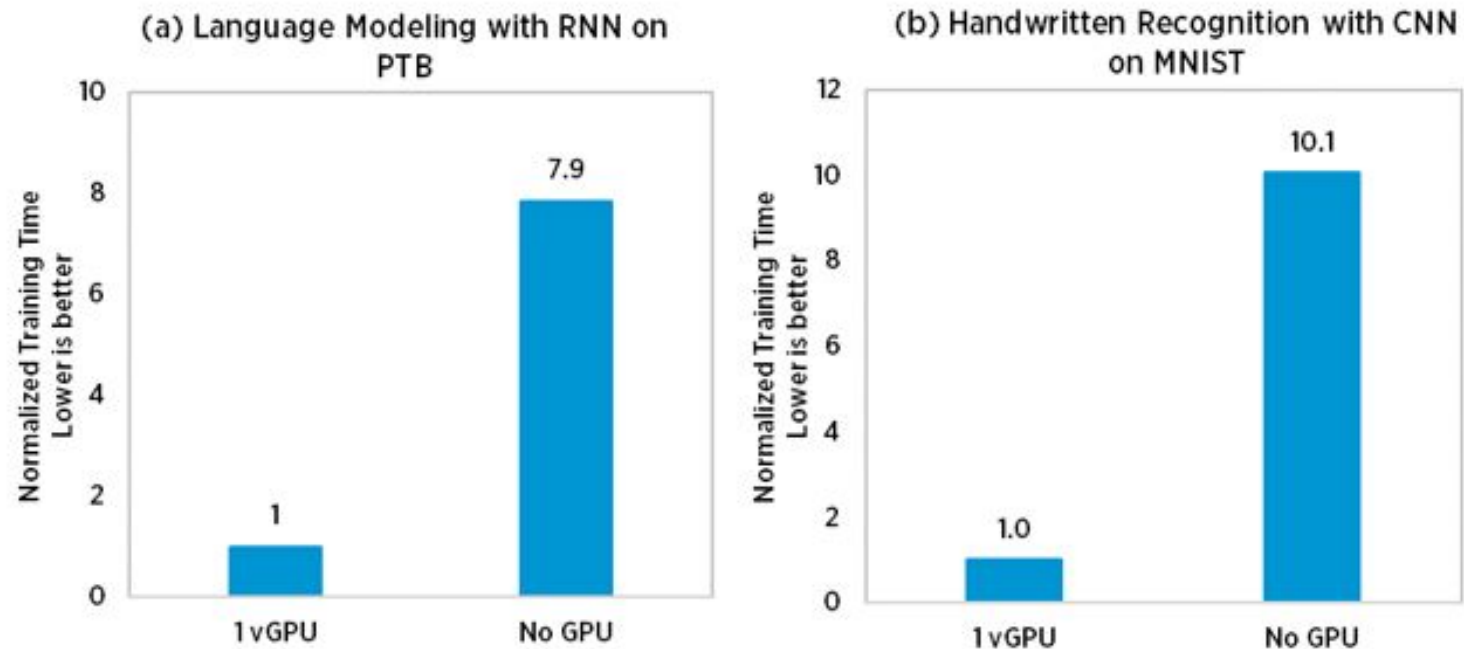
- GPUs contain thousands of Arithmetic Logic Units (ALUs) in a single processor
- The large number of processors means you can execute thousands of multiplications and additions simultaneously

Source: <https://cloud.google.com/tpu/docs/intro-to-tpu>

How GPU Help ?

- GPU architecture works well on applications with massive parallelism, such as matrix operations in a neural network.
- Benefits of using GPU for ML
 - Your machine learning operations will run faster than on CPU
 - Your training process will be distributed
 - You will work with a large memory bandwidth
 - You will be able to process large datasets faster. The larger the dataset the greater benefit you can gain from using GPUs
 - You will be able to use some advanced techniques to enhance the performance even more
- GPU is still a general-purpose processor that has to support many different applications and software
 - **For every calculation in the thousands of ALUs, a GPU must access registers or shared memory to read operands and store the intermediate calculation results**

How GPU Help ?



Source:
<https://blogs.vmware.com/performance/2017/10/episode-3-performance-comparison-native-gpu-virtualized-gpu-scalability-virtualized-gpus-machine-learning.html>

How to Use GPU

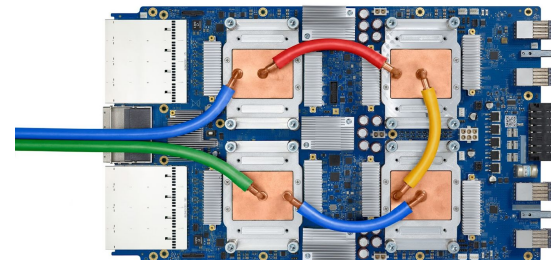
- CUDA is a parallel computing platform and programming model by [NVIDIA](#) that focuses on general computing on GPUs. CUDA speeds up various computations helping developers unlock the GPUs' full potential.
- The NVIDIA CUDA Deep Neural Network library or [cuDNN](#) is a GPU-accelerated library of primitives for deep neural networks. cuDNN provides highly tuned implementations for standard routines such as
 - forward and backward convolution, pooling, normalization, and activation layers.
- TensorFlow code, and [tf.keras](#) models will transparently run on a single GPU with no code changes required.

**When people ask you
how learning CUDA is going**

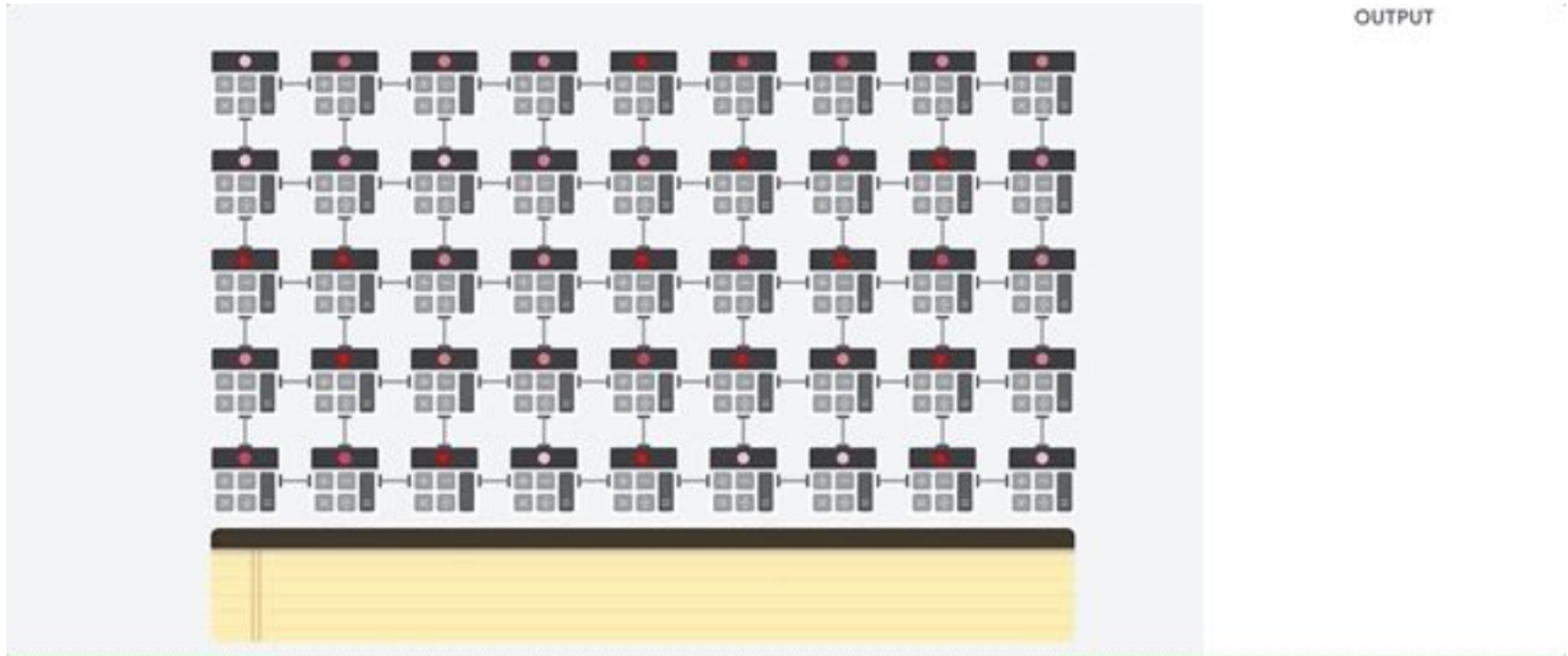


What is TPU

- Matrix processor specialized for neural network workloads.
- TPUs can't run word processors, control rocket engines, or execute bank transactions, but they can handle massive matrix operations used in neural networks at fast speeds.
- TPUs contain thousands of multiply-accumulators that are directly connected to each other to form a large physical matrix. This is called a systolic array architecture.



How TPU Works



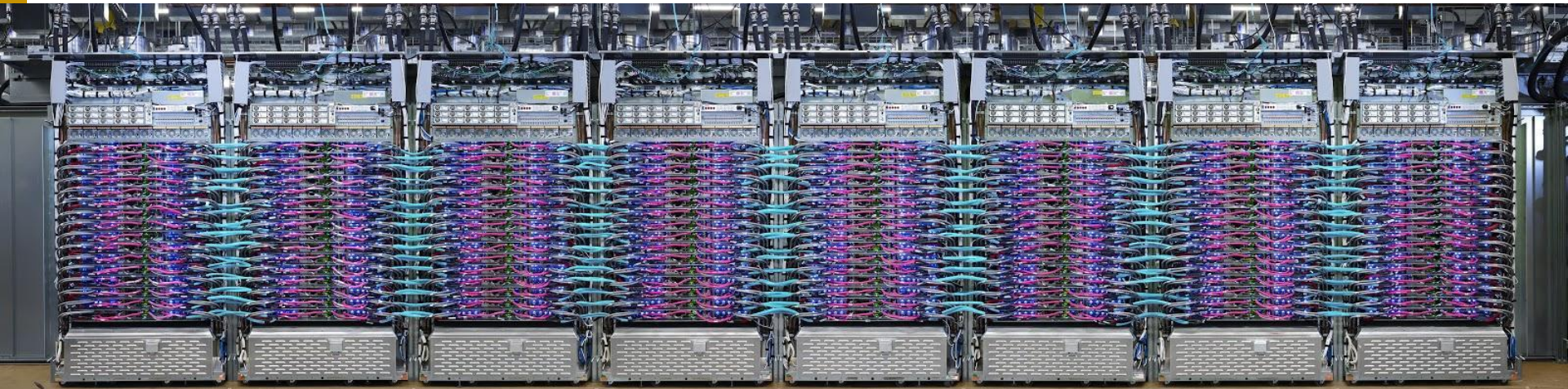
- TPU loads the parameters from memory into the matrix of multipliers and adders
- Then, the TPU loads data from memory
- As each multiplication is executed, the result will be passed to next multipliers while taking summation at the same time
- **No memory access is required during the matrix multiplication process**

Source: <https://cloud.google.com/tpu/docs/intro-to-tpu>

TPU Pods

A TPU Pod is a collection of TPU devices connected by dedicated high-speed network interfaces

A TPU Pod can have up to 2048 TPU cores, allowing you to distribute the processing load across multiple TPU



Lab

Lab Activities

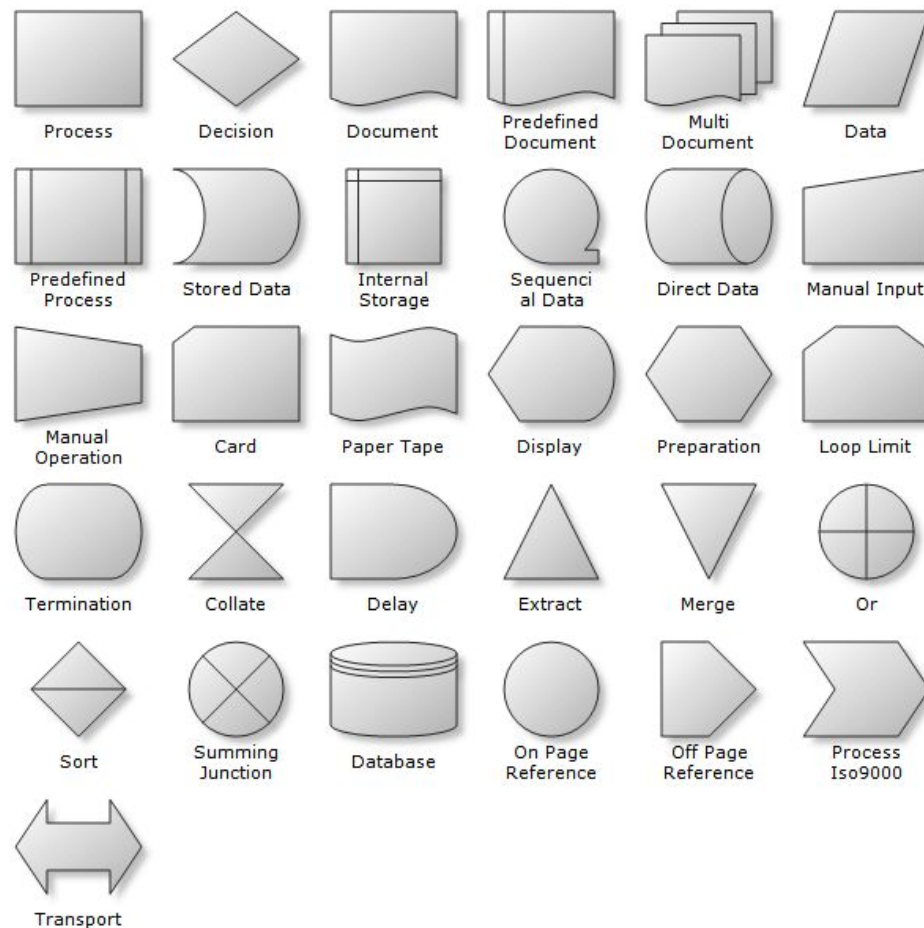
1. Build a .py script that takes a snapshot date, trains a model and outputs artefact into storage.

Dataset to be provided by instructor

original source datasets for instructors: <https://www.kaggle.com/datasets/parisrohan/credit-score-classification>

Assignment & Projects

Draw your system diagrams



An aerial night photograph of Singapore. The foreground shows the modern architecture of the Singapore Management University (SMU) campus, with its distinctive tiered roof and glass facades. The middle ground is filled with dense urban development and trees. The background features the iconic Singapore skyline, with numerous skyscrapers illuminated against the dark sky. The overall scene is a blend of modern education and a vibrant city.

Questions?

Thank You